

Dynamic sys.

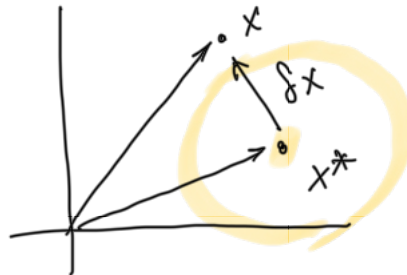
$$\begin{aligned}\dot{x} &= f(t, x, u) \\ y &= g(t, x, u)\end{aligned}$$

Equilibrium x^* fixed sol.

$$\dot{x}^* = f(t, x^*, u^*) = 0$$

Linearization

$$x = x^* + \delta x$$

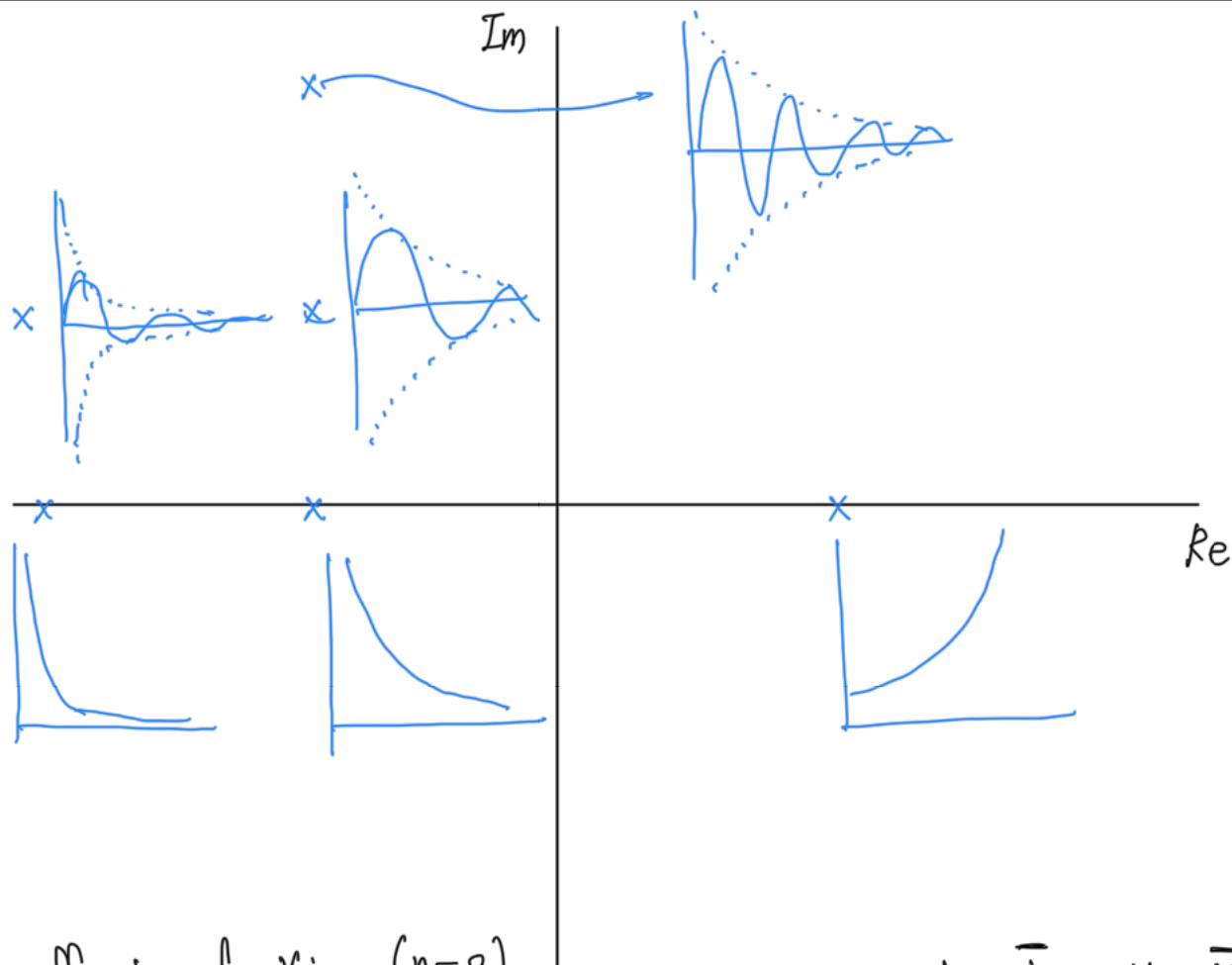


$$\|\delta x\| \ll 1$$

$$\begin{aligned}\delta \dot{x} &= A \delta x + B \delta u \\ \delta y &= C \delta x + D \delta u\end{aligned}$$

solution.

$$x(t) = \underbrace{e^{At} x_0}_{\text{zero-input resp. (} u=0 \text{)}} + \underbrace{\int_0^t e^{A(t-z)} B u(z) dz}_{\text{zero-state resp. (} x_0=0 \text{)}}$$



② effects of v_i ($n=2$)

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$$

$$\lambda_2 = \bar{\lambda}_1 \quad v_2 = \bar{v}_1$$

$$c_2 = \bar{c}_1$$

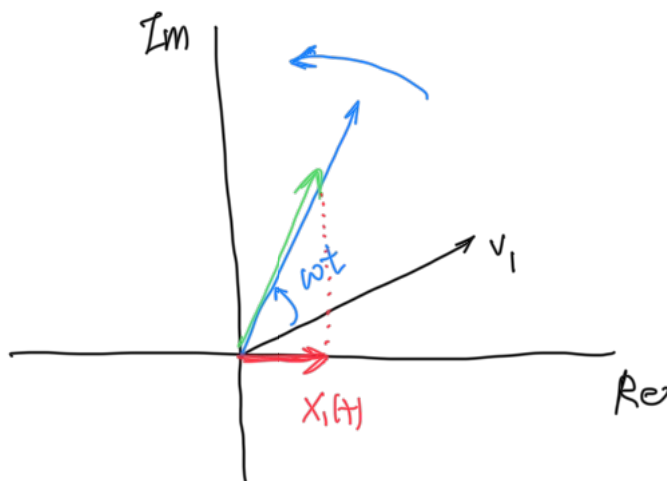
$$= 2 \operatorname{Re} [c e^{\lambda t} v]$$

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$x_1(t) = 2 \operatorname{Re} \left[c e^{\alpha t} e^{j\omega t} v_1 \right]$$

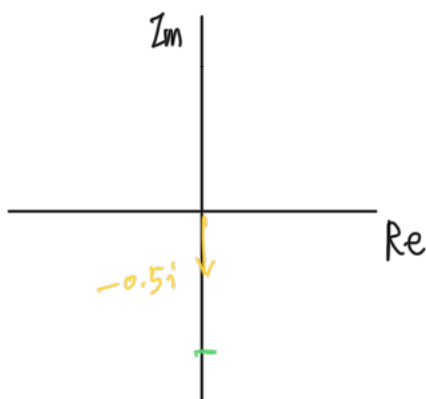
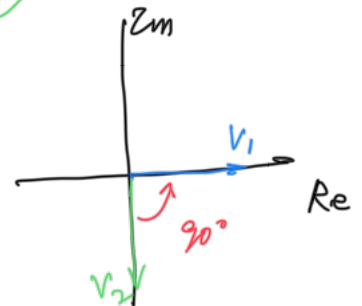
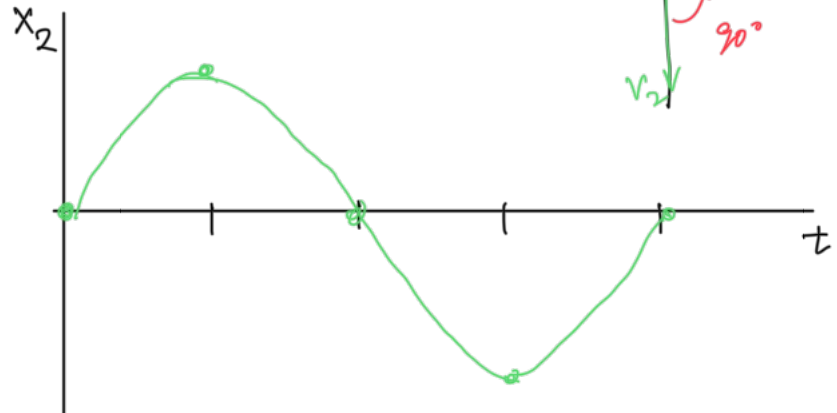
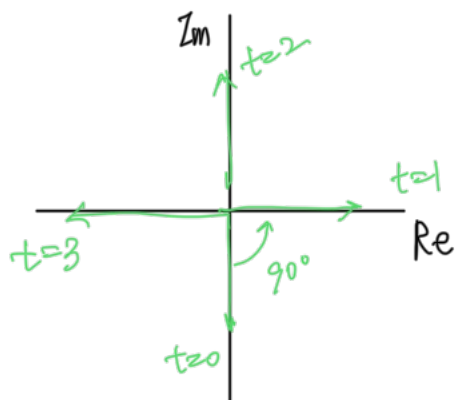
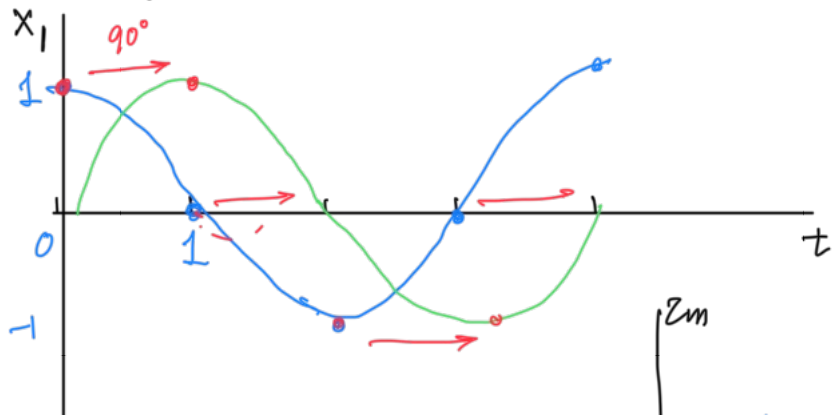
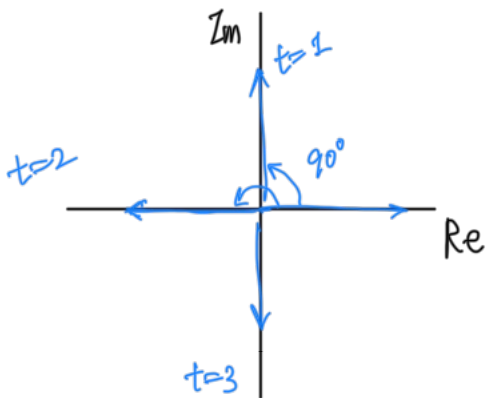
$$v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$c = \frac{1}{2}$$

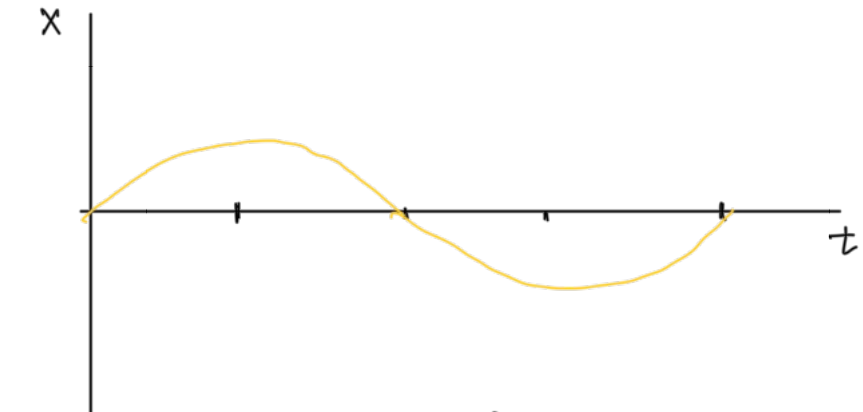


ex) $\lambda = \frac{\pi}{2}i$ $v = \begin{bmatrix} 1 \\ -i \end{bmatrix}$ $c = \frac{1}{2} \Rightarrow x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$
 $(\sigma = 0)$

$$x_1(t) = \operatorname{Re} \left[e^{i\frac{\pi}{2}t} \cdot \underline{1} \right] \quad x_2(t) = \operatorname{Re} \left[e^{i\frac{\pi}{2}t} \cdot \underline{(-i)} \right]$$

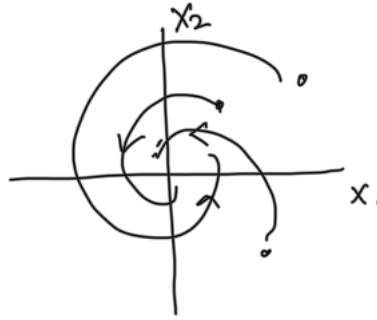
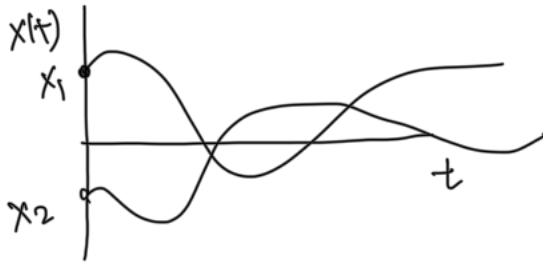


$$v = \begin{bmatrix} 1 \\ -0.5i \end{bmatrix}$$



(rel mag/phase $x(t)$)
 $\Leftrightarrow$ (a of e-rec)

Phase portrait



stability

$\left\{ \begin{array}{l} \text{a.s. } (\Leftrightarrow \text{e.s.}) \\ \text{stable (but not a.s.)} \\ \text{unstable} \end{array} \right. \quad \left\{ \begin{array}{l} \text{Re}[\lambda_i] < 0 \text{ for all } i \\ \text{Re}[\lambda_i] \leq 0 \text{ " } \\ \text{if } \text{Re}[\lambda_i] = 0, \text{ then } \lambda_i \text{ is semi-simple.} \\ \text{Re}[\lambda_i] > 0 \text{ for any } i \\ \text{or } \text{if } \text{Re}[\lambda_i] = 0, \text{ then } \lambda_i \text{ is not semi-simple.} \end{array} \right.$

ex) $A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$ $\lambda_1 = -1, -2.$ $\Rightarrow$ a.s.

$A = \begin{bmatrix} -1 & -3 \\ 3 & -1 \end{bmatrix}$ $\lambda = -1 \pm 3i$

$A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ $\lambda = \pm 2i$ $\text{Re}[\lambda] = 0$
 $\Rightarrow$ stable (not a.s.)

$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\lambda = 0, 0$ $\text{Re}[\lambda] = 0$
 (size of J) = 2. $\Rightarrow$ unstable.
 not semi-simple

< Controllability >

$$\underbrace{x(t)} = \underbrace{e^{At}} \cdot \underbrace{x_0} + \boxed{\int_0^t e^{A(t-\tau)} B u(\tau) d\tau}$$

Q: how can we change $x(t)$ using $u(t)$.

$$\boxed{} = \mathcal{C} \times (\text{func. of } u(t))$$

$$\mathcal{C} = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

if $\text{rank}[\mathcal{C}] = n$, $\boxed{}$ can take any value in $\mathbb{R}^n$.
 $\Rightarrow$ controllable.

if $\text{rank}[\mathcal{C}] = r < n$, $\boxed{}$ can take any value
 in the range of $\mathcal{C}$

ex) $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\mathcal{C} = [B \quad AB \quad A^2B] = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$(col_3) = 2(col_2) - (col_1)$$

$$\text{Range}[\mathcal{C}] = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$\Rightarrow$ we can control x_2 arbitrarily

x_1 and x_3 are affected by u in the same way

ex) $x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow x(t) = \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix}$ Yes $\begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix}$ Yes $\begin{bmatrix} 10 \\ 5 \\ 10 \end{bmatrix}$ Yes $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ No

u cannot change $x_1 - x_3 = z$

$$\dot{z} = \dot{x}_1 - \dot{x}_3 = (x_1 + x_3) - (x_2 + x_3) = x_1 - x_3 = z$$

$$\Rightarrow \dot{z} = z \quad \boxed{\text{no control}} \quad \underline{z(t) = z(0)e^t}$$

PBH test.

(left e-vec of A)

$$w = \begin{bmatrix} a \\ b \\ -a \end{bmatrix}$$

$$w_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$w_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$w_1^T B = [1 \ 0 \ -1] \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0$$

not controllable
along $w = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
 $(x_1 - x_3)$

$$w_2^T B = 1.$$

< Observability >

Q: can we find $x(0)$ from $u(t), y(t)$

$$\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{bmatrix} = \begin{bmatrix} c \\ cA \\ \vdots \\ cA^{n-1} \end{bmatrix} \underbrace{x_0}_{\text{Q}} + \underbrace{\begin{pmatrix} \text{func of } u \end{pmatrix}}_{\text{yellow highlight}}$$

if $x(0) \in \text{Null}[O]$ then $Ox_0 = 0$, we cannot determine x_0

if $\text{rank}[O] = n \Rightarrow \text{null}[O] = \emptyset \Rightarrow \text{observability empty.}$

ex) $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad c = [0 \ 1 \ 1]$

$$O = \begin{bmatrix} c \\ cA \\ cA^2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -2 & -3 \\ 0 & 4 & 9 \end{bmatrix}$$

$\text{rank}[O] = 2 \Rightarrow \text{not observable}$

$$\text{null}[O] = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$y(t) = Cx(t) = C e^{At} x_0 = \begin{bmatrix} 0 & e^{-2t} & e^{-3t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}$$

$$y(t) = e^{-2t} x_2(0) + e^{-3t} x_3(0)$$

↑ is not affected by $x_1(0)$ at all.

ex) $x(0) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \Rightarrow x(0) = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$

Both have the same output response.

PBH test. (e-vec of A) $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ $v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$C v_1 = 0$ $\rightarrow$ any initial state along v_1 does not contribute to y .

$C v_2 = 1$

$C v_3 = 1$

Midterm

part 1: in class open note

part 2: take-home python coding.

HW3 – 1(a)–(b), matrix A_1 **Eigenpairs and hand-drawn phase portrait**Consider $\dot{x} = A_1 x$, where

$$A_1 = \begin{bmatrix} -3 & 1 \\ 2 & -2 \end{bmatrix}.$$

$$\lambda_1 = -1$$

$$\lambda_2 = -4$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

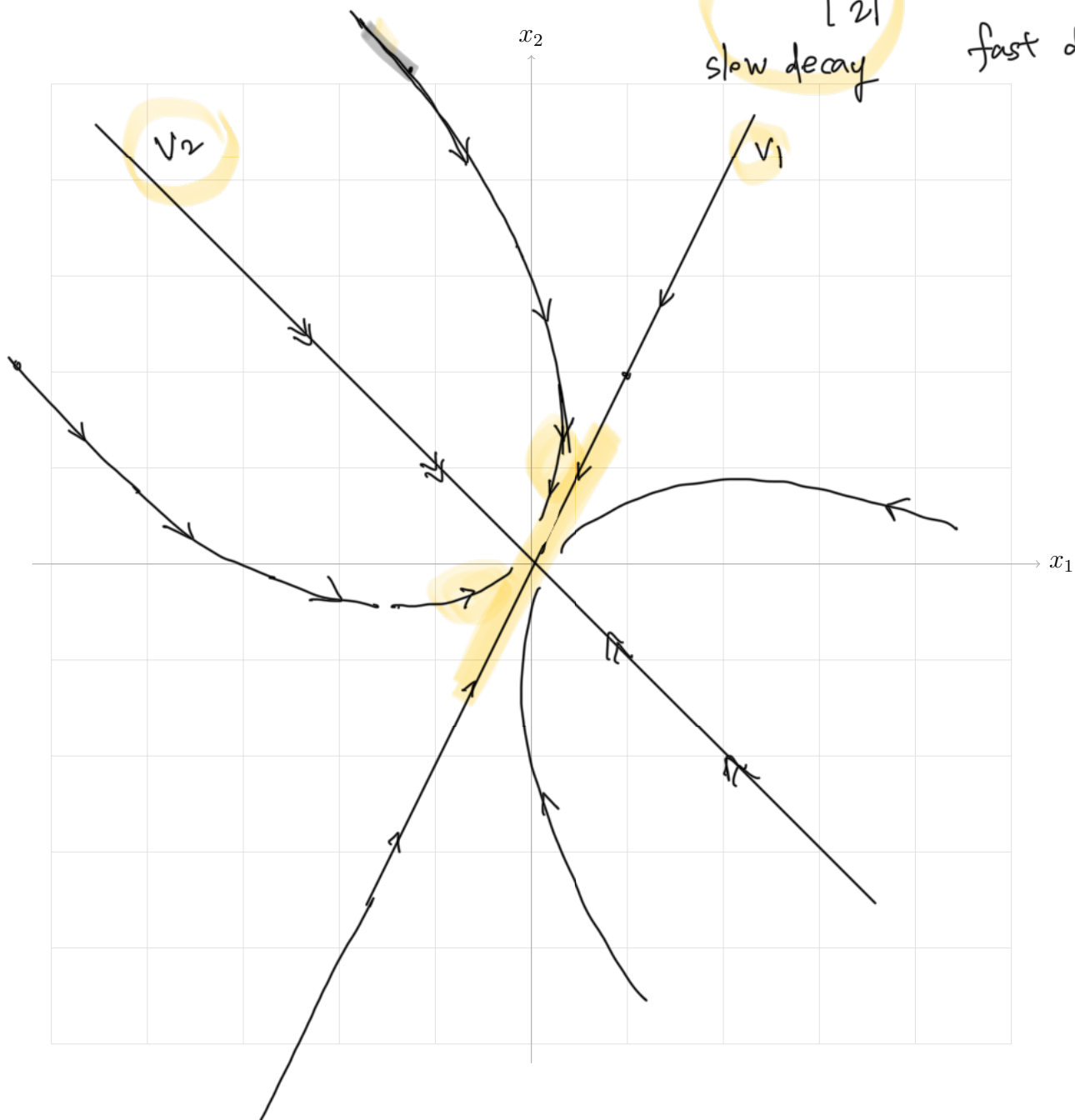
(a) Find the eigenvalues and corresponding eigenvectors analytically. Classify the origin as a stable node, unstable node, stable focus, unstable focus, center, or saddle.

(b) Sketch the phase portrait by hand. Include the origin, both eigenvector lines, representative curved trajectories, and arrows showing increasing time. Label the equilibrium type and briefly explain how the eigenpairs determine the sketch. No particular initial conditions are prescribed.

Class discussion and handwritten work

$$x(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

slow decay fast decay



HW3 – 1(a)–(b), matrix A_2 **Eigenpairs and hand-drawn phase portrait**

Consider $\dot{x} = A_2 x$, where

$$A_2 = \begin{bmatrix} -2 & 2 \\ 4 & 0 \end{bmatrix}.$$

$$\lambda_1 = 2 \quad \lambda_2 = -4$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

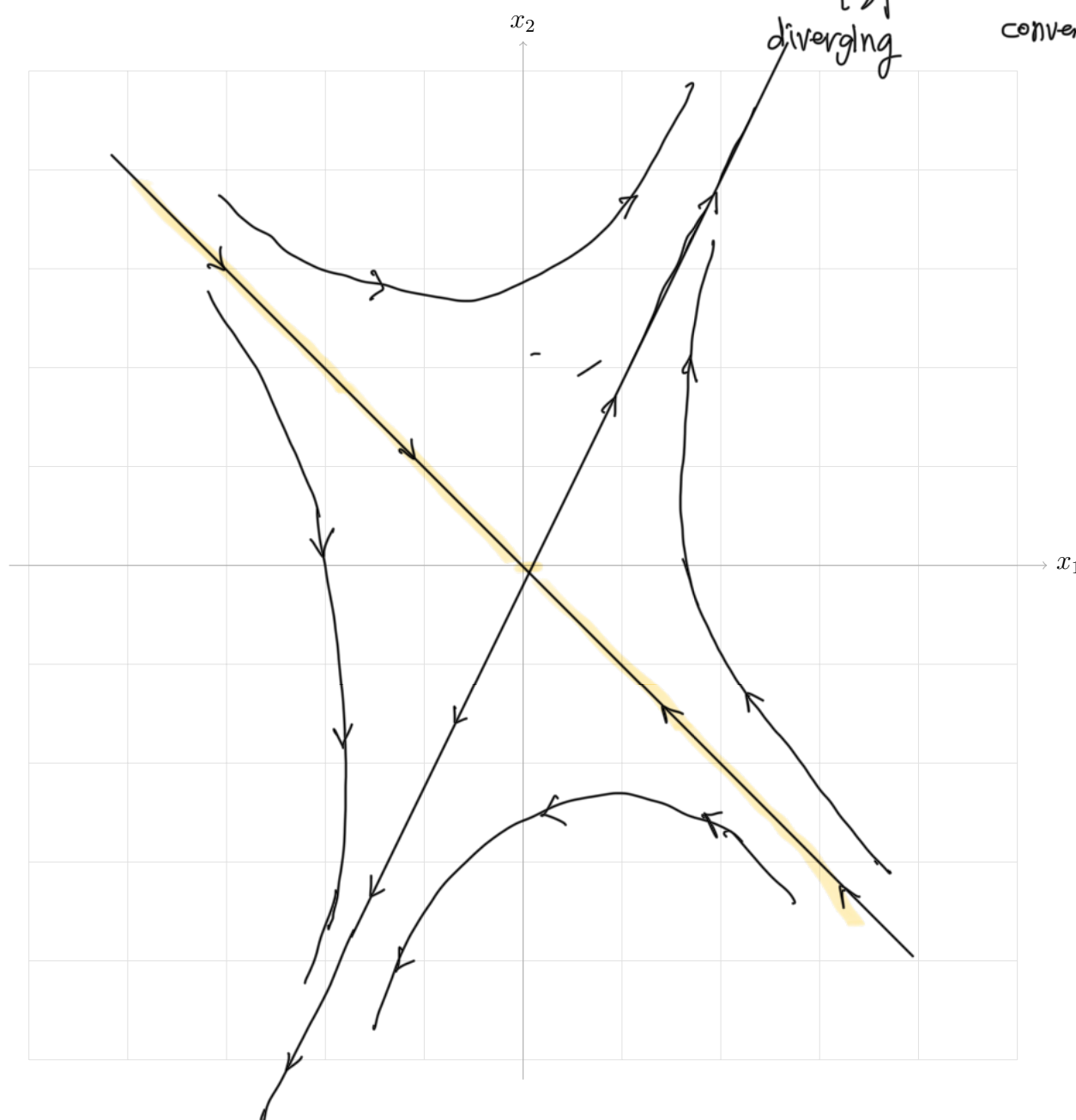
(a) Find the eigenvalues and corresponding eigenvectors analytically. Classify the origin as a stable node, unstable node, stable focus, unstable focus, center, or saddle.

(b) Sketch the phase portrait by hand. Include the origin, both eigenvector lines, representative curved trajectories, and arrows showing increasing time. Label the equilibrium type and briefly explain how the eigenpairs determine the sketch. No particular initial conditions are prescribed.

Class discussion and handwritten work

$$x(t) = c_1 e^{2t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

diverging converging



HW3 – 1(a)–(b), matrix A_3 **Eigenpairs and hand-drawn phase portrait**

Consider $\dot{x} = A_3 x$, where

$$A_3 = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}.$$

$$\lambda_1 = 1$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

slow div

$$\lambda_2 = 4$$

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

fast div

(a) Find the eigenvalues and corresponding eigenvectors analytically. Classify the origin as a stable node, unstable node, stable focus, unstable focus, center, or saddle.

(b) Sketch the phase portrait by hand. Include the origin, both eigenvector lines, representative curved trajectories, and arrows showing increasing time. Label the equilibrium type and briefly explain how the eigenpairs determine the sketch. No particular initial conditions are prescribed.

Class discussion and handwritten work

